

A P P E N D I X II

S U M M A R Y O F F O R M U L A S A N D
T H E I R C O N N E C T I O N T O S I N E X

This document gives a short summary of the basic formulas used for least squares adjustment and it gives instructions which vector or matrix of the normal equation system belongs to the individual SINEX blocks.

SUMMARY OF LEAST SQUARES ADJUSTMENT FORMULAS

You have n_{obs} linearized observation equations

$$(1) \quad v = A \, dx - l$$

where

n_{obs} number of observations
 v residual vector
 A Jacobian matrix
 dx corrections for the unknowns x concerning the apriori values x_0
i.e. $x = x_0 + dx$

with

n_{unk} number of unknowns
 l vector 'observed' minus 'computed with apriori values'.

P denotes the weight matrix for the observations.

The goal of least square adjustment is to minimize the square sum of residuals:

$$(2) \quad v' P v = \min$$

where v' is the transposed vector of v .

This condition leads to the so called normal equation

$$(3) \quad A' P A \, dx = A' P l$$

with normal equation matrix

$$(4) \quad N = A' P A$$

and the vector of the right hand side of the normal equation

$$(5) \quad b = A' P l .$$

The resulting unknown parameters can be determined with

$$(6) \quad x = x_0 + \text{inv}(A' P A) A' P l = x_0 + \text{inv}(N) b$$

where inv stands for the inverse matrix and x_0 are the apriori values.

The residuals can be computed with equation (1) and the aposteriori variance factor is then

$$(7) \quad s_0 = (v' P v) / (n_{\text{obs}} - n_{\text{unk}}) .$$

The weighted square sum of the vector l (= observed minus computed) can be obtained with

$$(8) \quad l' P l = v' P v + dx' b$$

$$= v' P v + dx' A' P l .$$

The variance-covariance matrix of the unknowns results in

$$(9) K = s_0 \text{ inv}(N).$$

If you introduce constraints as pseudo-observations with `n_constr` linearized observation equations

$$(10) v_c = H dx - h$$

with

`n_constr` number of constraints as pseudo-observations
`v_c` residuals over the constraints
`H` Jacobian matrix for pseudo-observation equations
`h` vector 'observed' minus 'computed' for the constraints.

`P_c` denotes the weight matrix for your pseudo-observations.

The least square methods lead to the normal equation for the pseudo-observations

$$(11) H' P_c H dx = H' P_c h$$

with normal equation matrix of constraints

$$(12) N_{\text{constr}} = H' P_c H$$

and vector of the right hand side of normal equation for constraints

$$(13) b_{\text{constr}} = H' P_c h.$$

The complete normal equation system for the constrained solution can easily be computed:

$$(14) (A' P A + H' P_c H) dx = A' P l + H' P_c h$$

with the constrained normal equation matrix

$$(15) N_{\text{total}} = A' P A + H' P_c H = N + N_{\text{constr}}$$

and the vector of the right hand side of the constrained normal equation system

$$(16) b_{\text{total}} = A' P l + H' P_c h = b + b_{\text{constr}} .$$

The unknown parameters of the constrained solution can be computed with

$$(17) x_c = x_0 + \text{inv}(N_{\text{total}}) b_{\text{total}} .$$

After computing the residuals over the constraints with equation (10) the weighted square sum of residuals of the constrained normal equation system can be obtained with

$$(18) v' P v + v_c' P_c v_c$$

and the number of degrees of freedom of the constrained normal equation system is

$$(19) \text{dof} = n_{\text{obs}} + n_{\text{constr}} - n_{\text{unk}} .$$

The aposteriori variance-factor for the constrained normal equation system is then

$$(20) s_{0_c} = (v' P v + v_c' P_c v_c) / \text{dof} .$$

The variance-covariance matrix for the unknowns of this constrained normal equation system can be computed with

$$(21) K_{xx} = s_{0_c} \text{ inv}(N_{\text{total}})$$

And the variance-covariance matrix for the constraints is

$$(22) K_{\text{constr}} = s_{0_c} \text{ inv}(N_{\text{constr}}) .$$

IMPLEMENTATION IN SINEX

The different elements belonging to the normal equations can be stored in SINEX files in the following way:

SOLUTION/STATISTICS block:

n_unk	=	NUMBER OF UNKNOWNNS
n_obs	=	NUMBER OF OBSERVATIONS
(20) s0_c	=	VARIANCE FACTOR
(18) $v' P v + v_c' P_c v_c$	=	SQUARE SUM OF RESIDUALS (VTPV)
(19) dof	=	NUMBER OF DEGREES OF FREEDOM

SOLUTION/ESTIMATE block:

(17) x_c in field "Parameter Estimate"

SOLUTION/APRIORI block:

x0 in field "Parameter Apriori"

SOLUTION/MATRIX_ESTIMATE block:

(21) Type COVA: K_xx
Type CORR: correlation matrix of K_xx
(15) Type INFO: N_total = N + N_constr

SOLUTION/MATRIX_APRIORI block:

(22) Type COVA: K_constr
Type CORR: correlation matrix of K_constr
(12) Type INFO: N_constr

SOLUTION/NORMAL_EQUATION_VECTOR block:

(5) $b = A' P l$

SOLUTION/NORMAL_EQUATION_MATRIX block:

(4) $N = A' P A$